

Hongyin Luo

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PARTICULARS

WORK

MIT Computer Science and Artificial Intelligence Laboratory Research Scientist	Cambridge MA, USA <i>July 2024 - Now</i>
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EDUCATION

Massachusetts Institute of Technology Ph. D. in Computer Science <i>Minor in Cognitive Science</i>	Cambridge MA, USA <i>May 2022</i>
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Massachusetts Institute of Technology S. M. in Computer Science	Cambridge MA, USA <i>May 2019</i>
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Tsinghua University B. Eng. in Computer Science and Technology	Beijing, China <i>June 2016</i>
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RESEARCH INTERESTS

I focus on AI reasoning, grounding, and efficiency. My research efforts attempt to improve structured reasoning, reduce hallucination, and eliminate redundant computation.

DISSERTATION

Title: “Self-training for Natural Language Processing”
Advisor: James R. Glass

My thesis develops a self-training framework to adapt efficient language models on different downstream tasks without using any human-annotated training data. This approach make small language models outperform 500 times larger GPT models on a wide range of standard natural language processing benchmarks.

Research Highlights

- Technique improves the reasoning capabilities of large language models. *MIT News*, June 2024. [\(link\)](#)
- IA générative: quelle technologie après ChatGPT? *Le Journal du Net*, July 2023. [\(link\)](#)
- The Cost of Doing AI Business. *PYMNTS*, July 2023. [\(link\)](#)

- Search engines don't always help chatbots generate accurate answers. *The Register*, June 2023. [\(link\)](#)
- Bigger is Not Always Better. *Analytics India Magazine*, June 2023. [\(link\)](#)
- MIT, 자가학습 AI 공개... 성능최대 500 배향상. *ZDNet Korea*, June 2023. [\(link\)](#)
- The Little Language Model That Could. *Hackster.io*, June 2023. [\(link\)](#)
- MIT researchers make language models scalable self-learners. *MIT News*, June 2023. [\(link\)](#)
- MIT researchers develop self-learning language models that outperform larger counterparts. *VentureBeats*, June 2023. [\(link\)](#)
- Текстовую нейросеть научили думать» еще лучше, чтобы избавить от расизма и сексизма. *Nauka TV*, March 2023. [\(link\)](#)
- Large language models are biased. Can logic help save them? *MIT News*, March 2023. [\(link\)](#)

WORK EXPERIENCE

- **Research Scientist, MIT CSAIL**, July 20224 - Now.
- **Postdoctoral Associate, MIT CSAIL**, July 2022 - June 2024.
- **Lecturer, MIT Momentum AI for minority and refugee**, September 2021 - January 2022.
- **Research Intern, Amazon AWS AI**, May 2021 - August 2021.
- **Research Intern, Google Brain**, May 2019 - August 2019.

TEACHING EXPERIENCE

- **MIT Kaufman Teaching Certificate**, Fall 2022, MIT Teaching + Learning Lab.
- **Lecturer**, Fall 2021, MIT Momentum AI.
- **Teaching Assistant**. 6.864 Advanced Natural Language Processing, Prof. Jacob Andreas and Prof. James R. Glass, Spring and Fall 2020, MIT.
- **Teaching Assistant and Project Mentor**. 6.862 Applied Machine Learning, Prof. Stefanie Jegelka, Spring 2019, MIT.

PUBLICATIONS

Preprints

1. Jiaxin Ge, Xueying Jia, Vijay Viswanathan, **Hongyin Luo**, Graham Neubig, "Training Task Experts through Retrieval Based Distillation," 2024.
2. Junmo Kang, Leonid Karlinsky, **Hongyin Luo**, Zhen Wang, Jacob Hansen, James Glass, David Cox, Rameswar Panda, Rogerio Feris, Alan Ritter, "Self-MoE: Towards Compositional Large Language Models with Self-Specialized Experts," 2024.
3. Tianhua Zhang, Kun Li, **Hongyin Luo**, Xixin Wu, James Glass, Helen Meng, "Adaptive Query Rewriting: Aligning Rewriters through Marginal Probability of Conversational Answers," 2024.
4. Philip Schroeder, Nathaniel Morgan, **Hongyin Luo**, James Glass, "THREAD: Thinking Deeper with Recursive Spawning", 2024.

PAPERS

5. Chi-Min Chan, Chunpu Xu, Ruibin Yuan, **Hongyin Luo**, Wei Xue, Yike Guo, Jie Fu, “Rq-rag: Learning to refine queries for retrieval augmented generation,” COLM 2024.
6. Zhaorun Chen, Zhuokai Zhao, **Hongyin Luo**, Huaxiu Yao, Bo Li, Jiawei Zhou, “HALC: Object Hallucination Reduction via Adaptive Focal-Contrast Decoding,” ICML 2024.
7. Junmo Kang, **Hongyin Luo**, Yada Zhu, James Glass, David Cox, Alan Ritter, Rogerio Feris, Leonid Karlinsky “Self-Specialization: Uncovering Latent Expertise within Large Language Models,” ACL 2024.
8. Tianhua Zhang*, Jiaxin Ge*, **Hongyin Luo***, Yung-Sung Chuang, Yuan Gong, Yoon Kim, Xixin Wu, Helen Meng, James Glass, “Natural Language Embedded Programs for Hybrid Language Symbolic Reasoning,” NAACL 2024.
9. Zhaorun Chen, Zhuokai Zhao, **Hongyin Luo**, Huaxiu Yao, Bo Li, Jiawei Zhou, “HALC: Object Hallucination Reduction via Adaptive Focal-Contrast Decoding”, ICLR 2024 R2-FM Workshop.
10. Yung-Sung Chuang, Yujia Xie, **Hongyin Luo**, Yoon Kim, James Glass, Pengcheng He, “Dola: Decoding by contrasting layers improves factuality in large language models,” ICLR 2024.
11. Yuan Gong, **Hongyin Luo**, Alexander H. Liu, Leonid Karlinsky, James Glass “Listen, Think, and Understand,” ASRU 2023.
12. **Hongyin Luo***, Tianhua Zhang*, Yung-Sung Chuang, Yuan Gong, Yoon Kim, Xixin Wu, Helen Meng, James Glass “Search-Augmented Instruction Learning,” EMNLP 2023.
13. Jiaxin Ge*, **Hongyin Luo***, Yoon Kim, James Glass, “Entailment as Robust Self-Learner,” ACL 2023.
14. Yuan Gong, Alexander H Liu, **Hongyin Luo**, Leonid Karlinsky, James Glass, “Joint Audio and Speech Understanding,” ASRU 2023.
15. Tianhua Zhang, Liping Tang, Wei Fang, **Hongyin Luo**, Xixin Wu, Helen Meng, James Glass, “ConvRGX: Recognition, Generation, and Extraction for Self-trained Conversational Question Answering,” Doc2dial Workshop at ACL 2023.
16. **Hongyin Luo**, James Glass, “Logic against bias: Textual entailment mitigates stereotypical sentence reasoning,” EACL 2023.
17. **Hongyin Luo**, Shang-Wen Li, Mingye Gao, Seunghak Yu, James Glass, “Cooperative Self-training of Machine Reading Comprehension,” NAACL 2022.
18. Yung-Sung Chuang, Rumen Dangovski, **Hongyin Luo**, Yang Zhang, Shiyu Chang, Marin Soljačić, Shang-Wen Li, Wen-tau Yih, Yoon Kim, James Glass, “DiffCSE: Difference-based contrastive learning for sentence embeddings,” NAACL 2022.
19. **Hongyin Luo**, James Glass, Garima Lalwani, Yi Zhang, Shang-Wen Li, “Joint Retrieval-Extraction Training for Evidence-Aware Dialog Response Selection,” EACL 2021.
20. Yung-Sung Chuang, Mingye Gao, **Hongyin Luo**, James Glass, Hung-yi Lee, Yun-Nung Chen, Shang-Wen Li, “Mitigating biases in toxic language detection through invariant rationalization,” WOAHI workshop at EMNLP 2021.

21. **Hongyin Luo**, Shang-Wen Li, James Glass, “Knowledge grounded conversational symptom detection with graph memory networks,” EMNLP 2021.
22. **Hongyin Luo**, Shuyan Dong, Yung-Sung Chuang, Shang-Wen Li, “Meta-learning for downstream aware and agnostic pretraining,” MetaNLP workshop at EMNLP 2020.
23. **Hongyin Luo**, Shang-Wen Li, James Glass, “Prototypical q networks for automatic conversational diagnosis and few-shot new disease adaption,” Interspeech 2020.
24. **Hongyin Luo**, Lan Jiang, Yonatan Belinkov, James Glass, “Improving neural language models by segmenting, attending, and predicting the future,” ACL 2019.
25. **Hongyin Luo**, Mitra Mohtarami, James Glass, Karthik Krishnamurthy, Brigitte Richardson, “Integrating Video Retrieval and Moment Detection in a Unified Corpus for Video Question Answering,” Interspeech 2019.
26. **Hongyin Luo**, James Glass, “Learning word representations with cross-sentence dependency for end-to-end co-reference resolution,” EMNLP 2018.
27. Jie Fu, **Hongyin Luo**, Jiashi Feng, Kian Hsiang Low, Tat-Seng Chua, “DrMAD: distilling reverse-mode automatic differentiation for optimizing hyperparameters of deep neural networks,” IJCAI 2016.
28. **Hongyin Luo**, Zhiyuan Liu, Huanbo Luan, Maosong Sun, “Online learning of interpretable word embeddings,” EMNLP 2015.

Software and Application

1. **LangCode**. A jupyter & Colab interface for generating natural language embedded program (NLEP).
2. **Anchoring**. A platform for generating large language model applications with natural language instructions, powered by neuro-symbolic augmented LLMs.
3. **SAIL-7b**. Search engine augmented large language model with entailment based retrieval denoising.
4. **UniLC**. A unified factual and moral language assessment toolkit.
5. **EntST**. Entailment self-training toolkit with simple pseudo-label editing (SimPLE) algorithm. Boosting small pretrained 350M-parameter language models to outperform GPT3-175B performance.
6. **RGX**. A framework for automatic question-answer data generation and self-training.

TALKS

INVITED TALKS

1. “Making Knowledge Agent-ready” *Boston Consulting Group*, Boston, May 2024
2. “Program Interfaces Grounded, Transparent, and Reasoning AI” *Chinese University of Hong Kong*, Hong King, March 2024
3. “Specialized Large Language Models”, *Quantiphi*, Cambridge MA, USA, October 2023.
4. “SAIL-7b: Search Augmented Instruction Learning LLM”,
 - *Sanofi*, Cambridge MA, USA, August 2023.

- *Amazon*, Sunnyvale CA, USA, July 2023.
 - *MassMutual*, Boston MA, USA, June 2023.
5. “Entailment as Robust Self-learner”, *IBM Watson AI Lab*, Cambridge MA, USA, May 2023.
 6. “Entailment Self-training”, *Embodied Intelligence Seminar*, Cambridge MA, USA, October 2022.
 7. “Cooperative Self-training for Machine Reading Comprehension”,
 - *MIT CSAIL Annual Meeting*, Cambridge MA, USA, June 2022.
 - *Embodied Intelligence Seminar*, Cambridge MA, USA, March 2022.
 - *Itaú Unibanco*, State of São Paulo, Brazil, February 2022.
 - *Chinese University of Hong Kong*, Hong Kong SAR, China, January 2022.
 - *Amazon Alexa*, Sunnyvale CA, USA, September 2021.
 8. “Belief State Generation for Misinformation Detection”, *Defence Science and Technology Agency (DSTA)*, Singapore, January 2021

CONFERENCE TALKS

9. “Search Augmented Instruction Learning”, *EMNLP 2023*, Singapore, December 2023.
10. “Entailment as Robust Self-learner”, *ACL 2023*, Toronto ON, Canada, July 2023.
11. “Logic Against Bias: Textual Entailment Mitigates Stereotypical Sentence Reasoning”, *EACL 2023*, Dubrovnik, Croatia, May. 2023.
12. “Cooperative Self-training of Machine Reading Comprehension”, *NAACL 2022*, Seattle QA, USA, July 2022.
13. “Knowledge Grounded Conversational Symptom Detection with Graph Memory Networks”, *EMNLP 2021*, Punta Cana, Dominican Republic, November 2021.
14. “Joint Retrieval-Extraction Training for Evidence-Aware Dialog Response Selection”, *Interspeech 2021*, Shanghai, China, May 2021.
15. “Prototypical Q Networks for Automatic Conversational Diagnosis and Few-shot New Disease Adaption”, *Interspeech 2020*, Virtual, October 2020.
16. “Integrating Video Retrieval and Moment Detection in a Unified Corpus for Video Question Answering”, *Interspeech 2019*, Graz, Austria, September 2019.
17. “Improving Neural Language Models by Segmenting, Attending, and Predicting the Future”, *ACL 2019*, Florence, Italy, July 2019.
18. “Learning Word Representations with Cross-sentence Dependency for End-to-end Co-reference Resolution”, *EMNLP 2018*, Brussels, Belgium, November 2018.
19. “Online Learning of Interpretable Word Embeddings”, *EMNLP 2015*, Lisbon, Portugal, September 2015.

PROFESSIONAL ACTIVITIES

- Reviewer (Journals) - *Springer Artificial Intelligence Review (AIRE)*, *ACL Rolling Review (ARR)*.
- Reviewer (Conferences) - *COLING* (2018, 2021, 2022), *Neurips* (2022), *ACL* (2018, 2019, 2020, 2021, 2022, 2023), *EMNLP* (2018, 2019, 2020, 2021, 2022, 2023), *NAACL* (2021, 2022), *EACL* (2022, 2023), *IJCAI-PRICAI* (2020), *WNGT* (2020), *PRMI* (2020), *WOHA* (2022, 2023).
- Mentor - *Tianhua Zhang* (Ph.D. student at Chinese University of Hong Kong), *Mark Razanau*, *Luc Gaitskell*, *Kimberly Wong* (undergraduate students at MIT), *Jiaxin Ge* (undergraduate student at Peking University).